

# **Enhancement of Temporal Sampling for Remote Sensing of Water Vapor Profiles, Clouds and Precipitation on a Global Basis: Recent Results from the Temporal Experiment for Storms and Tropical Systems (TEMPEST) Missions**

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The Temporal Experiment for Storms and Tropical Systems – Demonstration (TEMPEST-D) mission demonstrated the first global observations from a multi-frequency microwave radiometer on a CubeSat, operating for nearly three years from 2018 to 2021. This NASA Earth Venture Technology mission exceeded expectations for scientific data quality, instrument calibration, radiometer stability, and mission duration. TEMPEST-D brightness temperatures were validated using scientific and operational microwave sensors, including GPM/GMI and four MHS sensors, operating at similar frequencies to TEMPEST-D channels at 87, 164, 174, 178 and 181 GHz. TEMPEST-D performance was shown to be comparable to or better than much larger operational sensors, in calibration accuracy, precision, stability and instrument noise, during its nearly 3-year mission.

Based on the success of TEMPEST-D, a nearly identical TEMPEST sensor produced for risk reduction was delivered to the U.S. Space Force and integrated with NASA/JPL's Compact Ocean Wind Vector Radiometer (COWVR). COWVR and TEMPEST were launched on December 21, 2021, on STP-H8 for 3 years of operations on the ISS. COWVR and TEMPEST have performed coordinated observations of Earth's oceans and atmosphere from the ISS since January 7, 2022.

Atmospheric inversion techniques have been developed to retrieve water vapor altitude profiles, as well as single-layer cloud liquid water and cloud ice water, from TEMPEST brightness temperatures, using ECMWF Reanalysis v5 (ERA5) data as an initial guess. These retrievals are enhanced through the inclusion of geostationary infrared data from GOES-16 ABI channels, increasing the number of levels and reducing the error of water vapor retrieval, particularly in the upper troposphere.

Previous studies have validated the accuracy and precision of TEMPEST-D brightness temperatures using only clear-sky and non-precipitating oceanic observations. Recent studies extend the validation of TEMPEST-D observations with those from GPM-GMI over all sky conditions, including precipitating storms over both oceans and land. These studies demonstrated an average correlation coefficient of 0.9 between TEMPEST-D and GMI brightness temperature observations over tropical cyclones, as well as close agreement in detected image features, such as the size and location of the eye of the hurricanes. The similar quality of TEMPEST-D and GMI data allowed combining eye locations to improve temporal sampling frequency for tracking of tropical cyclones by about 2.5 times, as compared with GMI alone.

Recent studies extend the cross-validation of TEMPEST-D passive microwave observations with RainCube active microwave observations, both from CubeSats, to validation using GPM-DPR radar reflectivity profiles. Comparison of self-normalized TEMPEST-D brightness temperatures with inverted, vertically-integrated cumulative reflectivity from DPR yielded a correlation coefficient of 0.9. Such active/passive microwave combined observations provide a basis for development of surface rain rate estimates and retrieval of the vertical structure of precipitation from combined TEMPEST and DPR observations.